

%nprocshared=64
%mem=320GB
%chk=PbRhaC10C10T4.chk
opt=loose freq apfd/gen scrf=(solvent=water) pseudo=read scf=xqc

PbRhaC10C10T4

0 1

Pb	0.00344886	1.08458681	-0.54579077
O	-1.56101150	-1.85980559	-1.87498752
O	-0.98523406	2.84300009	0.81878959
O	0.21210081	-1.43303896	0.01011252
C	-1.78358016	-4.41279577	1.09810891
O	-3.01232145	-3.71042590	0.85311444
C	-3.08601352	-3.11361867	-0.42481562
C	-2.06999341	-1.95218556	-0.54698426
C	-0.91792588	-2.20097588	0.41791881
C	-0.59504359	-3.68563402	0.45696757
O	-1.73532074	0.78231206	1.06023181
C	-1.82527944	2.02655691	1.28987413
C	-3.00937394	2.52157298	2.08228076
C	-4.18226392	2.85787580	1.14972260
O	-4.41629177	1.76466791	0.23077261
C	-3.95648658	4.08603748	0.29385572
C	-5.13083301	4.40087083	-0.62233281
C	-4.88894211	5.63794948	-1.47622878
C	-6.05598948	5.96591184	-2.39880386
C	-5.80467366	7.20488825	-3.24889006
C	-4.99074232	0.66151613	0.71603363
O	-5.37408690	0.54482488	1.85833533
C	-5.09008771	-0.41098264	-0.33687801
C	-4.96602163	-1.80972649	0.28263666
O	-4.39923207	-2.72519046	-0.67135350
C	-1.87211913	-5.85984507	0.65424772
O	0.58987062	-3.93850296	1.18807363
C	-6.31383254	-2.35084580	0.72393234
C	-6.22195241	-3.63261174	1.53532448
C	-7.58511010	-4.15535276	1.96598537
C	-7.50195127	-5.44042964	2.78011146
C	-8.86916259	-5.95859548	3.20814148
O	0.74366858	-3.08176833	-4.38149049
O	2.54770797	2.06955572	-0.05813621
O	-0.88150701	-4.62276282	-2.64000137
C	2.07565002	-5.80897392	-1.66794087
O	1.87540668	-4.46726380	-1.21851366
C	1.79442058	-3.50773767	-2.25138807
C	1.30208380	-4.09827681	-3.57112001
C	0.24589958	-5.14987552	-3.31768289
C	0.85975712	-6.28602453	-2.47242125
O	1.41279747	0.67560462	1.23773411

C	2.47543086	1.32647189	0.93653577
C	3.66477312	1.14126473	1.85599811
C	4.95248742	0.91647907	1.08983402
O	4.73722106	-0.27800490	0.29084544
C	6.13921161	0.70804911	2.01298977
C	7.46031392	0.47738296	1.29377640
C	8.61897729	0.27383038	2.26044293
C	9.94977317	0.04145577	1.55636037
C	11.10378424	-0.16128798	2.52999826
C	5.16368979	-0.28327819	-0.97712641
O	5.74017311	0.63377917	-1.51342283
C	4.79412279	-1.58022775	-1.65194917
C	3.27992367	-1.73434246	-1.74387748
O	3.03485066	-2.91702851	-2.51169254
C	2.33765766	-6.66481164	-0.45361981
O	1.33908337	-7.34875269	-3.27484078
C	2.58885358	-0.53755324	-2.38609839
C	3.08524551	-0.16954655	-3.77768751
C	2.18047787	0.85713870	-4.44509158
C	2.64467755	1.25118951	-5.84063554
C	1.72954797	2.27847902	-6.49463065
H	2.88625196	-1.88258122	-0.73292489
H	5.13299621	1.74321429	0.40170902
H	-4.30595031	-1.75698843	1.15280937
H	-5.08097019	2.97931640	1.76057884
H	-1.65145637	-4.35753519	2.18332345
H	-2.85413459	-3.84756926	-1.20564683
H	-2.54345350	-1.00855049	-0.27211129
H	-1.23627688	-1.86641835	1.41080282
H	-0.46278752	-4.04429114	-0.56530015
H	-3.31472052	1.74665672	2.78464660
H	-2.73989022	3.42425449	2.63291156
H	-3.77340475	4.93075165	0.96694190
H	-3.04016824	3.95364732	-0.29128749
H	-5.32860346	3.54158455	-1.27265461
H	-6.03641787	4.54361783	-0.01831246
H	-4.68663168	6.49910657	-0.82551566
H	-3.98081583	5.49390135	-2.07664222
H	-6.25481582	5.10435493	-3.04844736
H	-6.96282219	6.10639424	-1.79735443
H	-4.92022449	7.07531052	-3.88069098
H	-5.63277760	8.08460970	-2.62061607
H	-6.05597837	-0.29959700	-0.84304466
H	-4.32484650	-0.23928152	-1.09320385
H	-0.93523408	-6.37286556	0.88563319
H	-2.03598935	-5.92620905	-0.42408306
H	-2.68447627	-6.37629903	1.16967751
H	1.26160970	-4.14428162	0.51802928
H	-6.81117111	-1.56904350	1.30751410
H	-6.92469646	-2.51165973	-0.17426718

H	-5.69795253	-4.39521807	0.95043393
H	-5.59999189	-3.45514538	2.42188102
H	-8.10637218	-3.38692418	2.55284763
H	-8.20752620	-4.32870947	1.07766601
H	-6.98249135	-6.20666575	2.19101415
H	-6.87760358	-5.26611920	3.66544902
H	-9.49992749	-6.16846570	2.33840703
H	-9.39424481	-5.22087396	3.82326427
H	2.94889691	-5.83428167	-2.33232424
H	1.08136809	-2.75899263	-1.89652241
H	2.14573554	-4.57792889	-4.08369126
H	-0.06646182	-5.57250503	-4.28183045
H	0.08046529	-6.63433001	-1.78364778
H	3.76916277	2.05081123	2.45652490
H	3.47219654	0.31222041	2.53940306
H	6.21539312	1.59671787	2.65043639
H	5.92420190	-0.13552447	2.68047567
H	7.37810448	-0.40038418	0.64227132
H	7.67308933	1.32720156	0.63486285
H	8.70388991	1.14827503	2.91923920
H	8.40165394	-0.57892875	2.91741466
H	9.86161858	-0.83208189	0.89834981
H	10.16375930	0.89393319	0.89964596
H	10.92527413	-1.02633505	3.17656774
H	11.22952866	0.71199700	3.17787877
H	5.23661686	-1.58639946	-2.64809286
H	5.20664962	-2.41572191	-1.08071404
H	2.45283758	-7.70743848	-0.75514484
H	1.50186771	-6.59950057	0.24775819
H	3.24892526	-6.34756975	0.05744556
H	0.58563567	-7.82044780	-3.64386134
H	1.51618884	-0.76276438	-2.43370167
H	2.70069416	0.32765341	-1.72301114
H	4.10352824	0.22915510	-3.71531226
H	3.15678489	-1.06804489	-4.40393478
H	1.15681174	0.46245888	-4.49939457
H	2.12200868	1.75329356	-3.81315857
H	3.66617177	1.64689425	-5.78248598
H	2.70318023	0.35281370	-6.46779881
H	1.67929926	3.19534002	-5.89884389
H	0.70969440	1.89234123	-6.58937620
H	1.42180658	-2.41467323	-4.53470034
H	-0.98686859	-3.68408139	-2.85433227
H	0.84217573	-1.43156270	0.74403231
H	-2.25380768	-1.49741436	-2.43739188
C	12.43716352	-0.39409097	1.79548994
H	12.65968314	0.45437355	1.18271513
H	12.35843859	-1.26676458	1.18137012
C	2.21687300	2.66268465	-7.90406328
H	3.19973935	3.08073555	-7.83997600

H	2.23993189	1.79055015	-8.52353863
C	-6.99715446	7.51600744	-4.17235873
H	-7.87185099	7.68662336	-3.58016266
H	-7.16629103	6.68741428	-4.82789377
C	-8.75821257	-7.25699087	4.02878726
H	-8.28268999	-8.01235435	3.43869890
H	-8.17797636	-7.07410708	4.90900284
C	13.56358535	-0.59205094	2.82679286
H	14.49059308	-0.75053217	2.31645928
H	13.64012739	0.27931269	3.44304504
H	13.34267897	-1.44247693	3.43742860
C	-10.16676586	-7.73144294	4.43186648
H	-10.75058950	-7.90351259	3.55184262
H	-10.09038582	-8.63964293	4.99243660
H	-10.63799201	-6.98082488	5.03138166
C	-6.68729553	8.77412173	-5.00466629
H	-7.52111290	8.99787670	-5.63679152
H	-6.50524429	9.59973365	-4.34882919
H	-5.82023786	8.59889880	-5.60666955
C	1.25641755	3.70043094	-8.51415177
H	1.25291784	4.58252916	-7.90851947
H	1.58057368	3.94820954	-9.50330725
H	0.26843194	3.29158530	-8.55452106

C H O 0
6-311+g(2d,p)

Pb	0	
S	4	1.00
	591.61124370	0.22126521076D-03
	46.757232559	0.56961959130D-02
	20.746462696	-0.21374063831
	14.610796419	0.40502620616
S	2	1.00
	20.222637612	-0.83541883299D-01
	6.4767324865	0.97910892388
S	1	1.00
	1.6600600927	1.0000000
S	1	1.00
	0.80431655001	1.0000000
S	1	1.00
	0.22627039020	1.0000000
S	1	1.00
	0.84014530665D-01	1.0000000
P	3	1.00
	15.189102118	0.61952303583
	14.693144415	-0.72498497086
	6.8705890048	0.37680007984
P	3	1.00
	2.2021426123	0.40196284806

	1.2209125119	0.46058131862
	0.63367559815	0.19367655397
P	1 1.00	
	0.28202837058	1.0000000
P	1 1.00	
	0.11333375666	1.0000000
P	1 1.00	
	0.43948707430D-01	1.0000000
D	6 1.00	
	61.315369628	0.33870800787D-03
	12.372195840	0.13788683942D-01
	6.9254944983	-0.75979608103D-01
	2.3319539939	0.28113784298
	1.2108730003	0.44474512269
	0.60090478506	0.35326874351
D	1 1.00	
	0.28135869813	1.0000000
D	1 1.00	
	0.11500000000	1.0000000
F	1 1.00	
	1.0000000	1.0000000
F	1 1.00	
	0.2896200	1.0000000

PB 0

PB-ECP 3 60

f potential

2

2	3.88751200	12.20989200
2	3.81196300	16.19029100

s-f potential

4

2	12.29630300	281.28549900
2	8.63263400	62.52021700
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

p-f potential

6

2	10.24179000	72.27689700
2	8.92417600	144.59108300
2	6.58134200	4.75869300
2	6.25540300	9.94062100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100

d-f potential

6

2	7.75433600	35.84850700
2	7.72028100	53.72434200
2	4.97026400	10.11525600

2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100